

# Application of the PACE Method

Redesign of Production Indicator by Stages (SQL Server → SAP HANA)

## 1. Introduction

This document describes the application of the PACE method (Plan, Analyze, Construct, Evaluate) to redesign and migrate a complex stored procedure from SQL Server to SAP HANA. The objective is a structural reengineering focused on performance, scalability, and maintainability.

## 2. Plan / Prepare

A large monolithic stored procedure (900+ lines) with heavy use of temporary tables and duplicated logic was identified. The goal was to redesign the solution into a modular SAP HANA-compatible architecture.

## 3. Analyze

The functional flow and technical issues were evaluated, including excessive UNION operations, scalar subqueries, and risks of NULL propagation and division errors.

## 4. Construct

The solution was rebuilt using SQLScript with modular functions, dynamic parameterization, and integration with SAP tables. Hardcoded logic and dependencies were eliminated.

## 5. Evaluate

Results showed a 40–60% reduction in code complexity, improved execution performance, and better maintainability.

## 6. Conclusion

The PACE methodology enabled a successful transformation into a scalable and optimized SAP HANA solution aligned with best practices.

## 7. Final Implementation

The final system integrates dynamic calculations, error handling mechanisms, and real-time metrics, ensuring robust and consistent performance across all stages.